

◆ Question 1: Positive & Negative Numbers

```
nums = [4, -2, 7, -5, 0, 3, -1]
```

Write a program to:

1. Store positive numbers in a **list**.
 2. Store negative numbers in a **set**.
 3. Count zeros using loop.
 4. Create a **dictionary**:
 5. {
 6. "positive": count,
 7. "negative": count,
 8. "zero": count
 9. }
 10. Print result as a **tuple**: (positive_count, negative_count, zero_count).
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◆ Question 2: Find Maximum & Minimum

```
nums = [12, 5, 8, 20, 3, 15]
```

Write a program to:

1. Find maximum using **for loop** (don't use `max()`).
 2. Find minimum using loop.
 3. Store them in a **tuple**.
 4. Create a **dictionary**:
 5. {
 6. "max": value,
 7. "min": value
 8. }
 9. Use if-else comparison.
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◆ Question 3: Count Vowels & Consonants

```
word = "education"
```

Write a program to:

1. Store vowels in a **set**.
 2. Store consonants in a **list**.
 3. Create a **dictionary** for character frequency.
 4. Use if-else inside loop.
 5. Print result as tuple: (vowel_count, consonant_count).
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◆ Question 4: Sum of Even & Odd

```
nums = [1, 2, 3, 4, 5, 6]
```

Write a program to:

1. Find sum of even numbers.
 2. Find sum of odd numbers.
 3. Store even numbers in a list.
 4. Store odd numbers in a set.
 5. Create dictionary:
 6. {
 7. "even_sum": value,
 8. "odd_sum": value
 9. }
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◆ Question 5: Duplicate Elements

```
nums = [1, 2, 3, 2, 4, 5, 1, 6]
```

Write a program to:

1. Find duplicate elements using loop.
2. Store duplicates in a **set**.
3. Store unique elements in a **list**.
4. Create dictionary:
5. {
6. element: frequency
7. }
8. Print result as tuple: (total_unique, total_duplicates).